



Timeline-Based Prediction of Chronic Diseases from Electronic Health Records

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Motivation

- Chronic diseases (diabetes, hypertension, heart disease) develop silently over time.
- Existing ML models predict if a diseases occurs but **not** when.
- This limits timely clinical intervention.
- Longitudinal EHR data offers an opportunity to predict personalized disease onset timelines, improving early intervention and resource allocation.

Introduction

A Multi-disease timeline prediction framework using synthetic EHR data (Synthea) targeting **diabetes, hypertension, and heart disease** simultaneously.

Pipeline Overview:

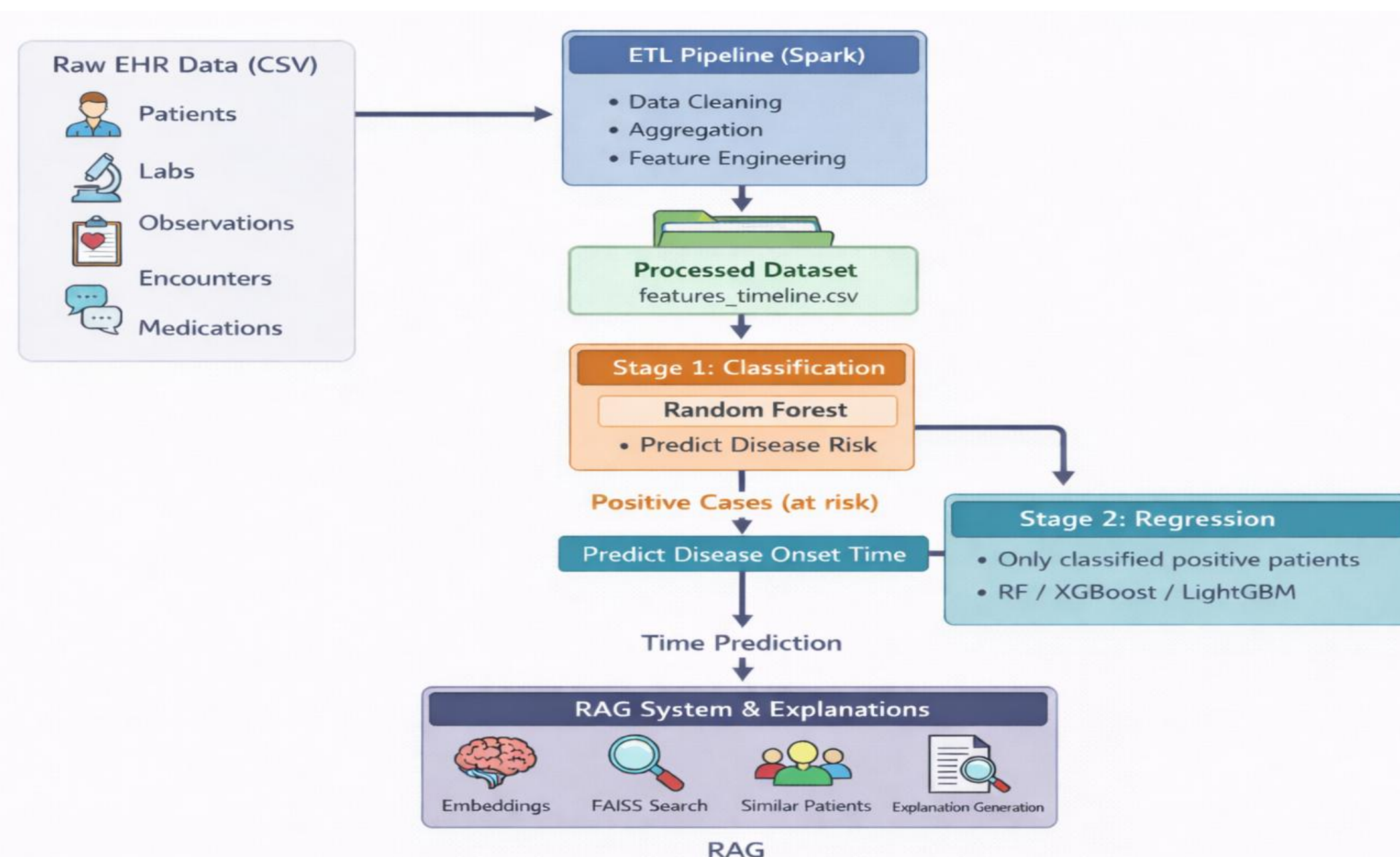
- PySpark ETL transforms raw EHR into patient-level features.
- Cutoff date (Jan 1, 2019) prevents data leakage.
- Two-stage ML: Classify risk → estimate days to onset.
- RAG layer explains predictions via similar patient retrieval.

Features used: Demographics · Lab values (LOINC-coded) · Clinical history · Time-to-event targets

LOINC Codes Used

Laboratory Test	LOINC Code
Hemoglobin A1c (HbA1c)	4548-4
Fasting Blood Glucose	1558-6
Random Blood Glucose	2339-0
Total Cholesterol	2093-3
Systolic Blood Pressure	8480-6
Diastolic Blood Pressure	8462-4

System Architecture



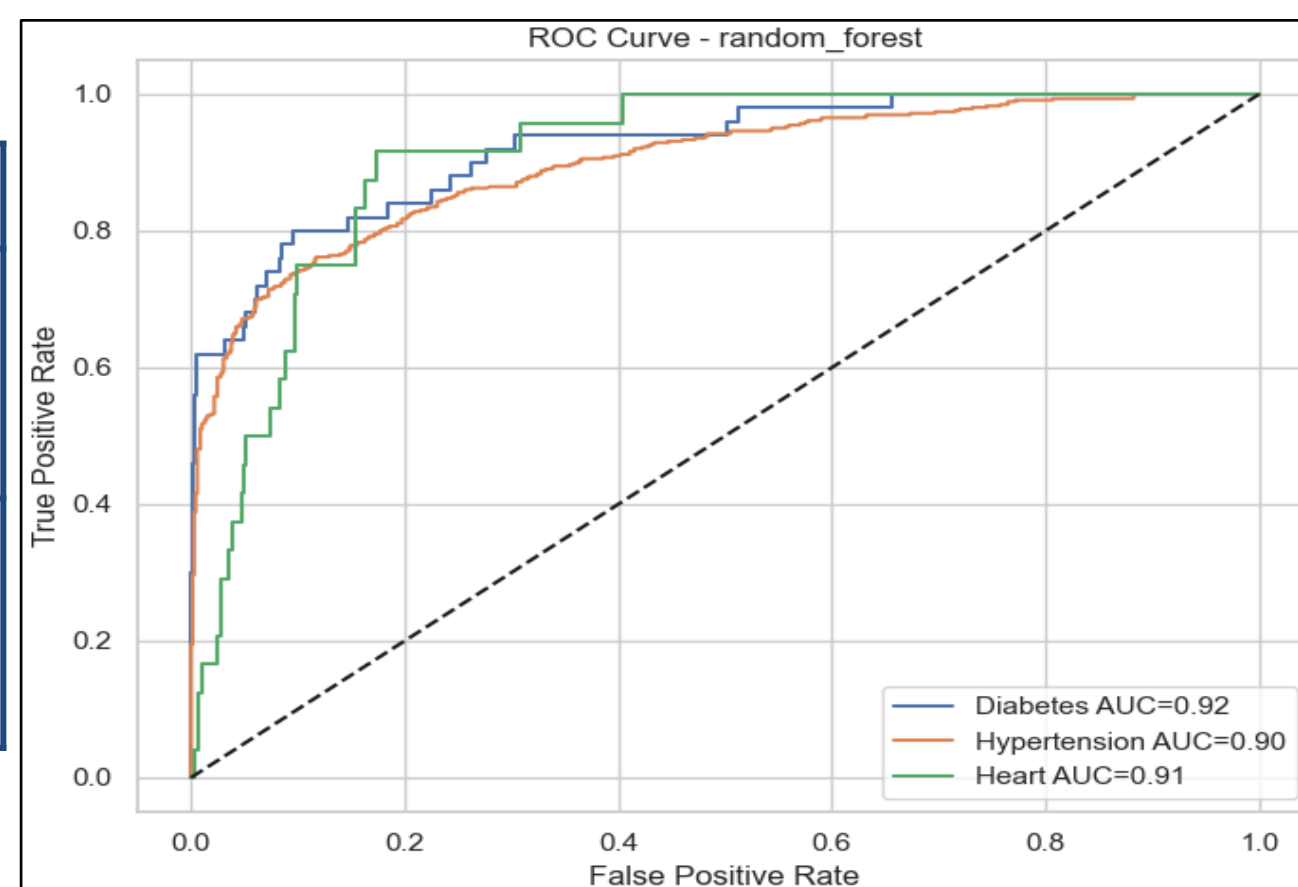
Experimental Setup

Settings	Detail
Data	Synthea synthetic EHR
Split	80% train / 20% test
Imbalance	SMOTE
Classification Metric	Accuracy, ROC-AUC
Regression Metric	MAE (days)
Retrieval Metrics	Precision@K, Recall@K, MMR
Embeddings	Sentence Transformer (MiniLM) + FAISS

Experimental Results

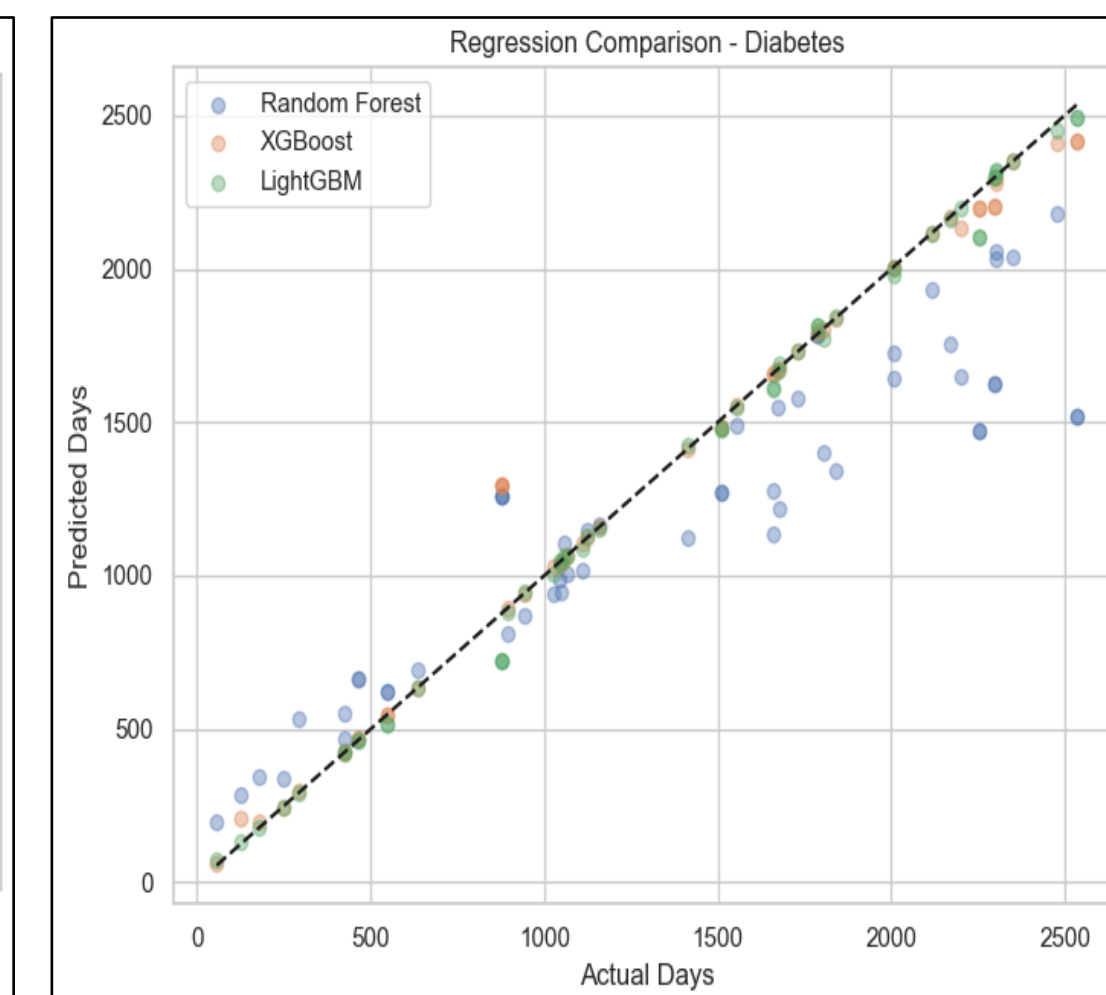
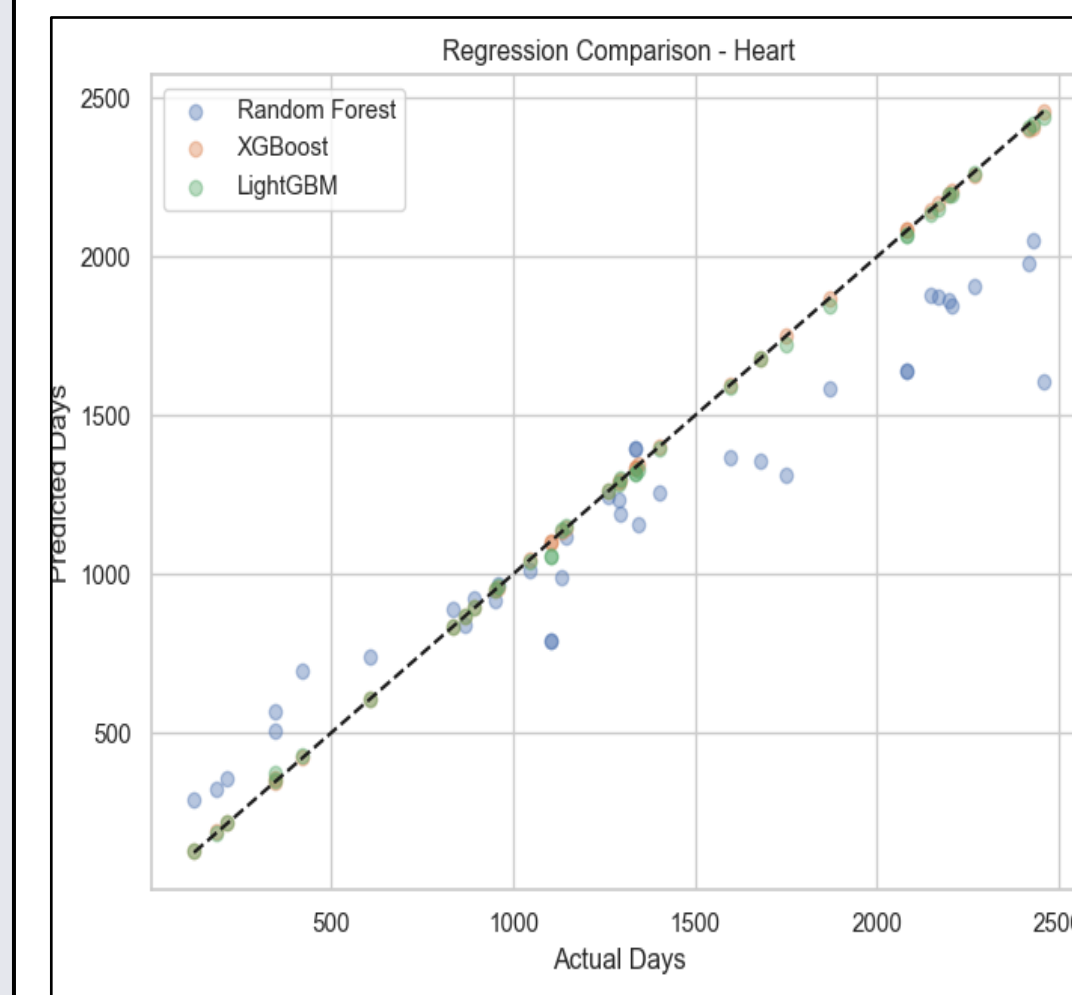
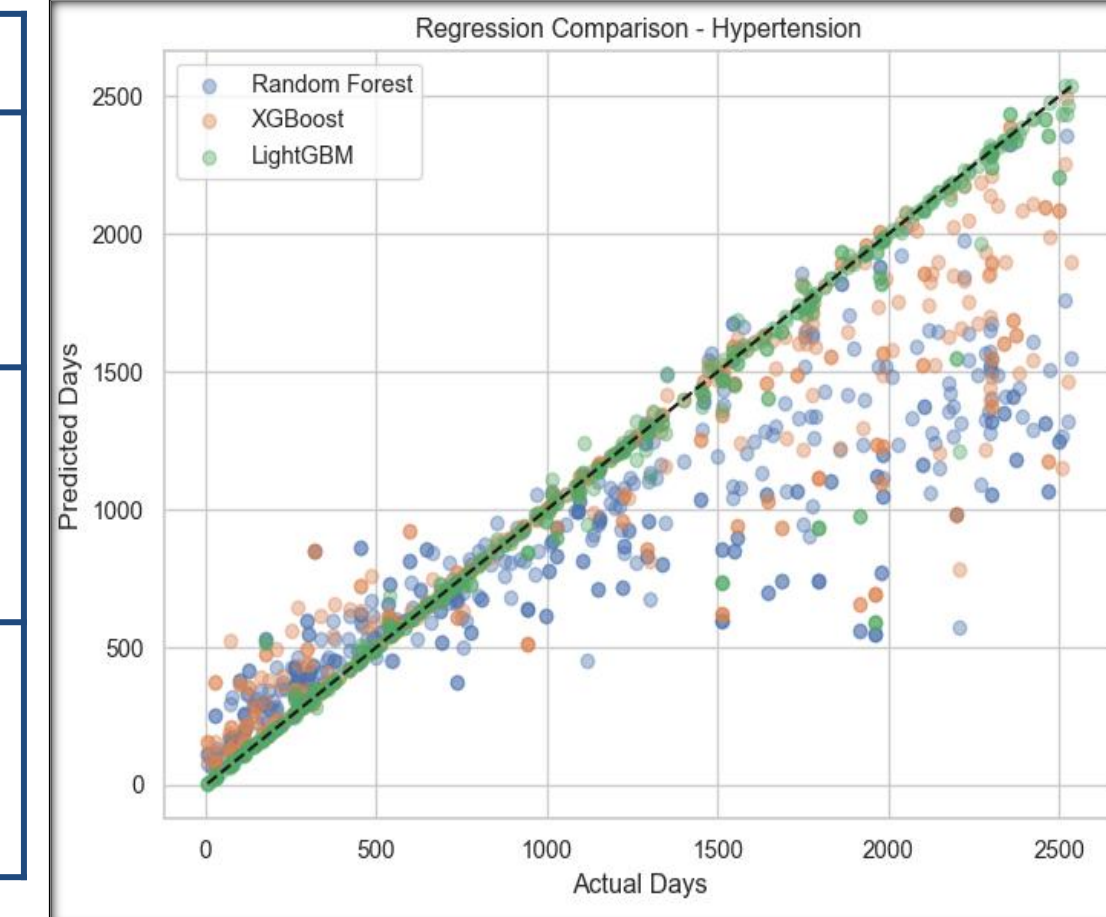
Classification Results

Model	Diseases	ACC	AUC
Logistic Regression	Diabetes	0.79	0.88
	Hypertension	0.74	0.84
	Heart Disease	0.75	0.84
Random Forest	Diabetes	0.92	0.92
	Hypertension	0.90	0.90
	Heart Disease	0.91	0.90



Regression Results

Model	Diseases	MAE
Random Forest	Diabetes	275.24
	Hypertension	349.11
	Heart Disease	217.82
XGBoost	Diabetes	41.19
	Hypertension	194.53
	Heart Disease	30.2
LightGBM	Diabetes	27.25
	Hypertension	53.13
	Heart Disease	18.18



Conclusion & Future Work

This project developed an end-to-end pipeline for multi-disease prediction using EHR data, combining risk classification, time-to-onset estimation, and a RAG-based explainability layer. The RAG system achieved **0.80 Precision@5 and Recall@5**, demonstrating effective retrieval of similar patients and improved interpretability.

Future work includes survival models, medical-domain embeddings, Extend to additional diseases, and clinical deployment.

References

- H. Tsai, T.W. Yang, T.Y. Wu, Y.C. Tu, C.L. Chen, and C.F. Chou, "Multitask learning multimodal network for chronic disease prediction"
- L. Garza, A. Kotal, M. A. Grasso, and E. Umucu, "Retrieval-Augmented Framework for LLM-Based Clinical Decision Support"